

COP2073C Programming Assignment 3

Instructions

Write an R script which does the following:

1. Create a data frame from the data shown in the attached table. Use the provided column headings as the column names. Ensure the LastName column is not treated as a factor by using the appropriate option.

EmployeeID	LastName	Salary
1001	Smith	45000
1002	Norris	51000
1003	Johnson	36000
1004	Hart	39000
1005	Owens	42000

2. Run the summary() function to display a summary of the data frame.
3. Use the cbind() function to add a new column to your data frame named Exempt, which indicates whether the employee is exempt from overtime. Use subsetting and a logical condition to calculate the values. Any employee with a salary over 50,000 should be exempt ("Y"), while others should be non-exempt ("N").

Hint: Create a vector with the two possible values ("Y" and "N"). Use subsetting to assign the correct value by evaluating whether each employee's salary is over 50,000. Keep in mind that logical conditions return TRUE (1) or FALSE (0), so you'll need to add 1 to the logical result in order to correctly index into your vector of "Y" and "N".

4. Display the updated data frame
5. Extract and display the first two rows of your data frame.
6. Use the str() function to display the structure of the data frame, paying attention to how the Exempt column is stored..
7. Add an additional row for the following employee using rbind:

EmployeeID: 1006

LastName: Rogers

Salary: 55000

Exempt: Y

8. Display the final version of your data frame